Expt 9

Adaptive Filtering for Echo Cancellation

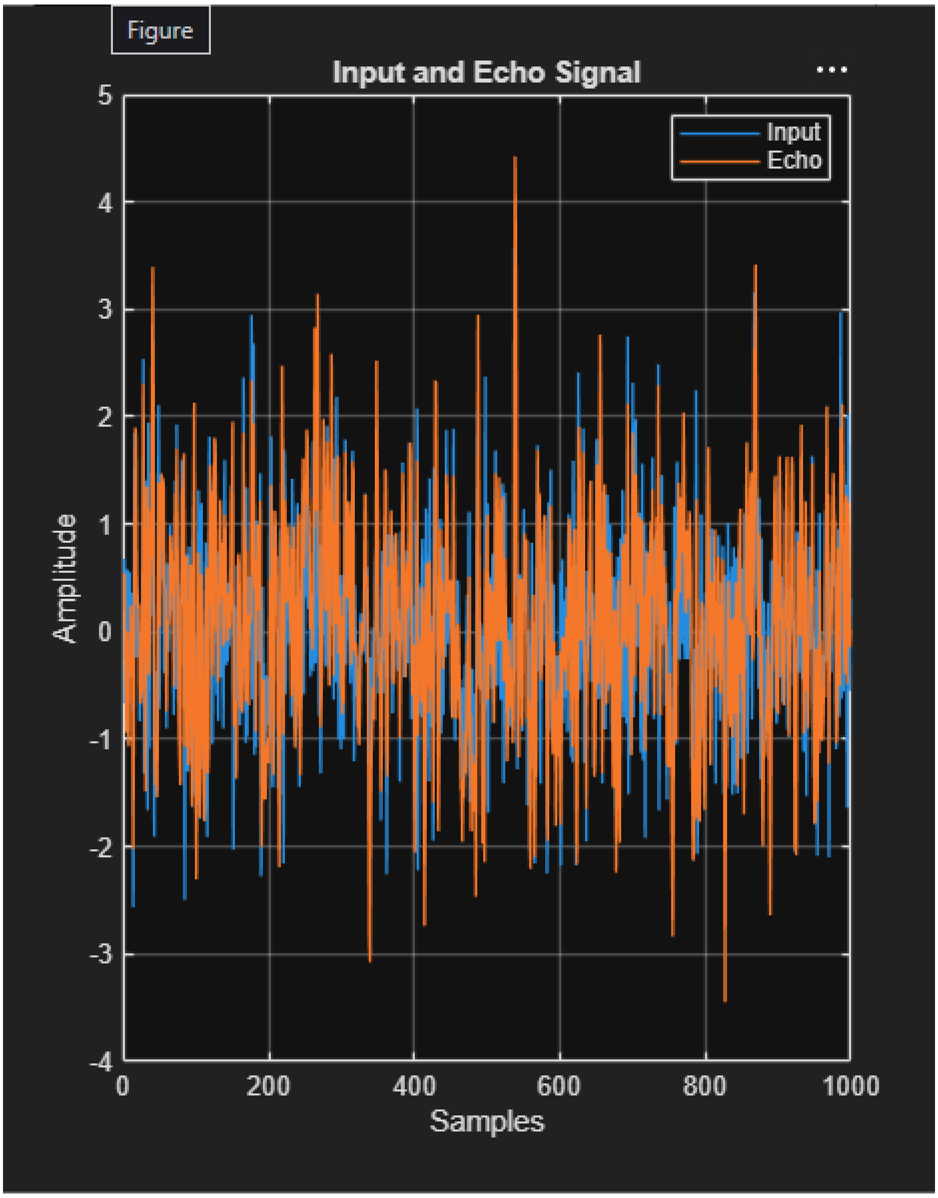
9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation:

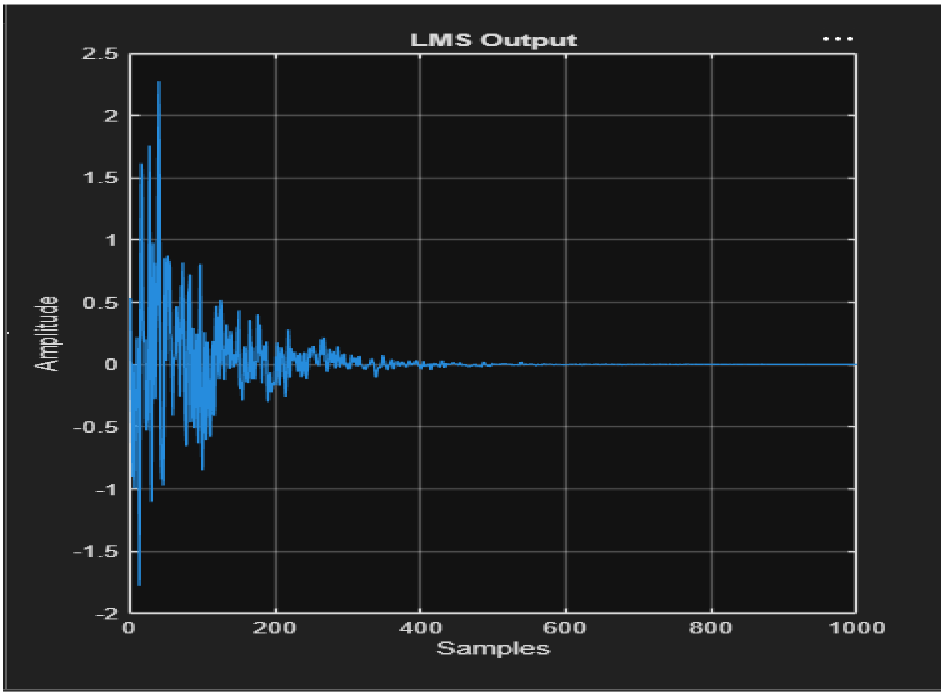
Program:

```
clc;
clear;
close all;

% Generate input signal
x = randn(1,1000);

% Generate echo signal
d = filter([0.8 0.5 0.3], 1, x);

% Exponential decay signals
e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));

% Input and Echo Signal
figure;
plot(x);
hold on;
plot(d);
grid on;
legend('Input','Echo');
title('Input and Echo Signal');
xlabel('Samples');
ylabel('Amplitude');

% LMS Output
figure;
plot(e1);
grid on;
title('LMS Output');
xlabel('Samples');
ylabel('Amplitude');

% NLMS Output
figure;
plot(e2);
grid on;
title('NLMS Output');
xlabel('Samples');
ylabel('Amplitude');

% MSE Comparison
```

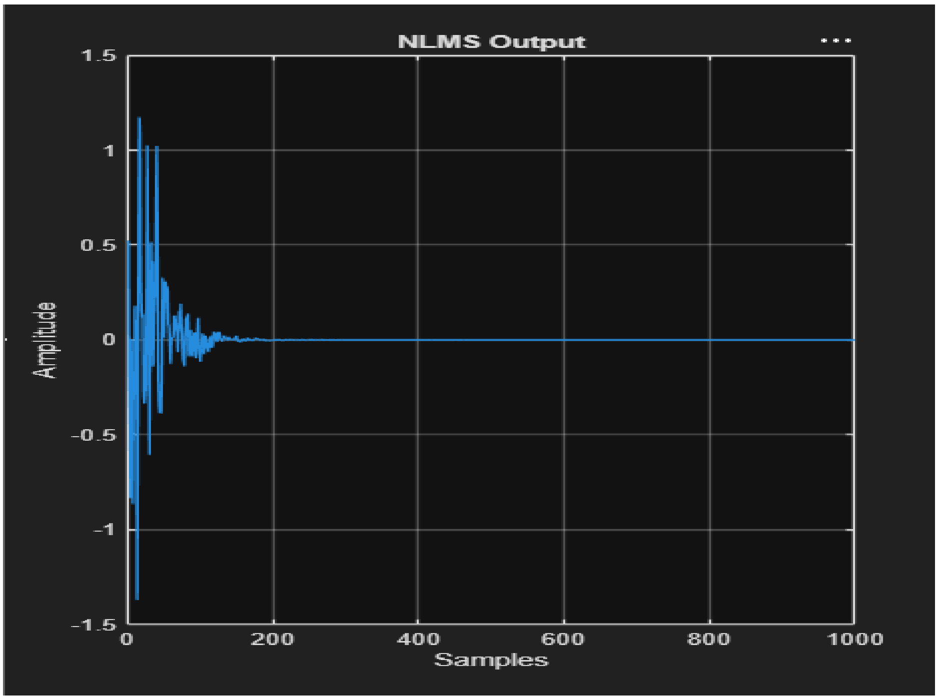
```
figure;
semilogy(e1.^2,'r');
hold on;
semilogy(e2.^2,'b');
grid on;
legend('LMS','NLMS');
title('MSE Comparison');
xlabel('Samples');
ylabel('Squared Error');
```

Output :

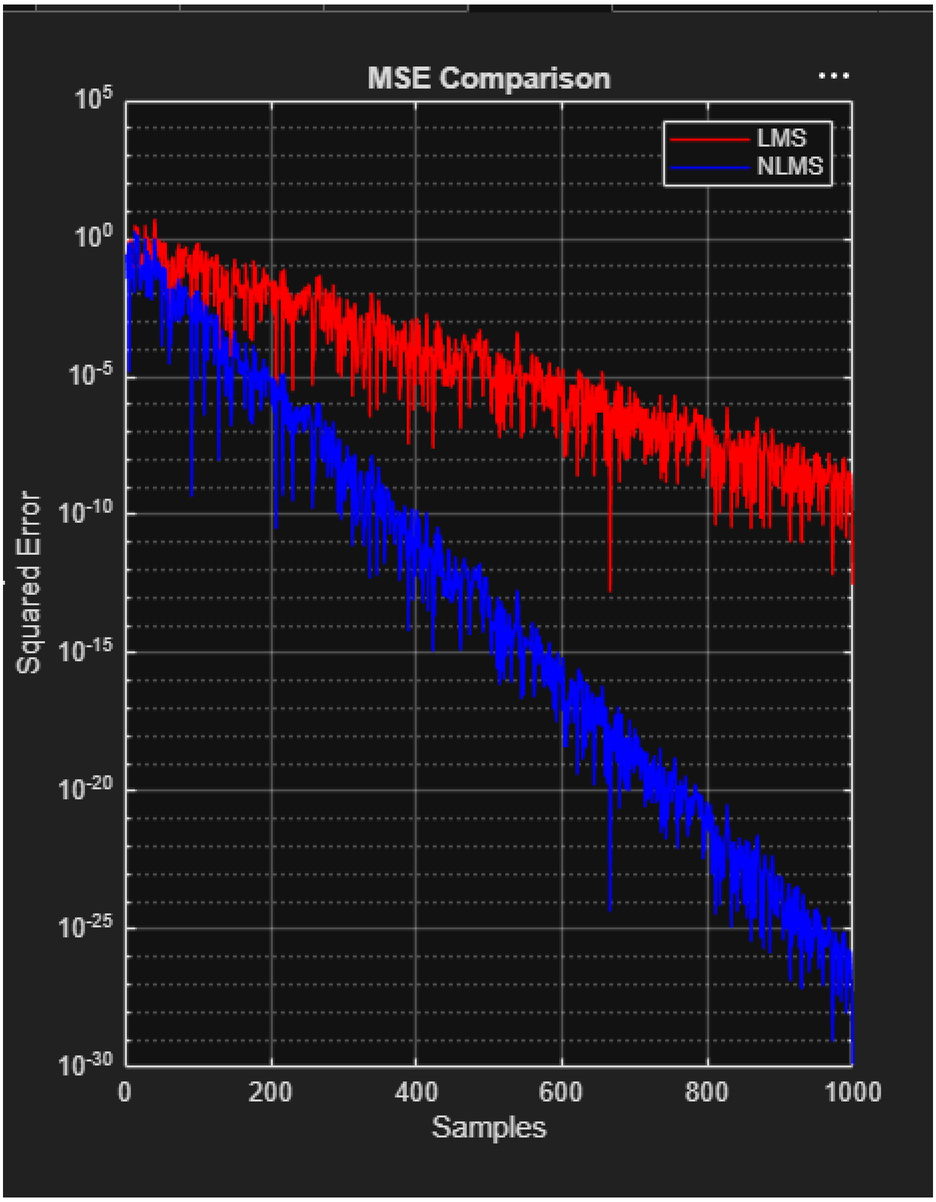




LMS Output
Amplitude
Samples
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
200
400
600
800
1000



NLMS Output
Amplitude
Samples
1.5
1
0.5
0
-0.5
-1
-1.5
0
200
400
600
800
1000


MSE Comparison
LMS
NLMS
Squared Error
Samples
10^5
10^0
10^-5
10^-10
10^-15
10^-20
10^-25
10^-30
0
200
400
600
800
1000